

Danny Wang

The University of Queensland, St Lucia, QLD 4072, Australia

danny.wang@uq.edu.au

[dannyw618.github.io](https://github.com/dannyw618)

ORCID 0009-0000-2783-8012

PhD candidate in machine learning at The University of Queensland, working on the reliability of models that meet the open world: out-of-distribution detection in graphs and text-rich networks, and controllable generation with discrete diffusion language models. My work has appeared at ICML, ICLR, EMNLP and KDD, including an ICLR spotlight in the top 5.1% of submissions.

EDUCATION

Doctor of Philosophy — Computer Science

01/2024 — 06/2027 (expected)

The University of Queensland, School of Electrical Engineering and Computer Science — Brisbane, Australia

Advisors: Dr Ruihong Qiu, Prof. Zi (Helen) Huang.

Bachelor of Computer Science / Master of Data Science

02/2020 — 11/2023

The University of Queensland, School of Electrical Engineering and Computer Science — Brisbane, Australia

Advisor: A/Prof. Tong (Rocky) Chen. GPA 6.83 / 7.0.

Class of 2023 Valedictorian; Dean's Award 2020, 2021, 2022, 2023.

PUBLICATIONS

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) 2026 — ACCEPTED

CORE A*

Wang, D., Qiu, R., and Huang, Z. (2026). What Information Matters? Graph Out-of-Distribution Detection via Tri-Component Information Decomposition.

UNDER REVIEW

Wang, D., Qiu, R., and Huang, Z. (2026). When to Commit? Towards Variable-Size Self-Contained Blocks for Discrete Diffusion Language Models.

EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP) 2025

MAIN CONFERENCE

CORE A*

Wang, D., Qiu, R., Bai, G., and Huang, Z. (2025). Text Meets Topology: Rethinking Out-of-Distribution Detection in Text-Rich Networks.

INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS (ICLR) 2025

SPOTLIGHT · TOP 5.1%

CORE A*

Wang, D., Qiu, R., Bai, G., and Huang, Z. (2025). GOLD: Graph Out-of-Distribution Detection via Implicit Adversarial Latent Generation.

KNOWLEDGE DISCOVERY AND DATA MINING (KDD) 2024

CORE A*

Chen, T., **Wang, D.**, Liang, X., Risius, M., Demartini, G., and Yin, H. (2024). Hate Speech Detection with Generalizable Target-aware Fairness.

AUSTRALASIAN JOINT CONFERENCE ON ARTIFICIAL INTELLIGENCE 2023

Australasian B

Wang, D., Wainwright, R., Layton, H., and Bialkowski, A. (2023). Beyond Model Accuracy: Identifying Hidden Underlying Issues in Chest X-ray Classification.

RESEARCH EXPERIENCE

Research Assistant — Fairness of Online Extremism Detection 02/2023 — 01/2024
The University of Queensland — supervised by Dr Rocky Chen (EECS) and Dr Marten Risius (BEL)

Developed and implemented a bias-mitigation method to improve trust and fairness in extremism-detection models, continuing the summer internship project. The resulting paper was published at KDD 2024.

Research Assistant — Driver Drowsiness Analysis 07/2022 — 09/2022
The University of Queensland — supervised by Dr Ruihong Qiu

Wrangled and analysed heart-rate variability, EEG and ECG signals, then applied machine-learning and deep-learning methods to identify heart-rate peaks and their correlation with driver drowsiness.

Summer Research Intern — UQ AI Collaboratory 11/2022 — 02/2023
On the Fairness of Data-Driven Online Extremism Detection — Dr Rocky Tong Chen, Dr Marten Risius

Studied fairness of extremism-detection models across advantaged and disadvantaged online groups. Processed and analysed over 100 million Reddit messages, and used BERT and BiRNN language models to surface bias in existing detection methods.

Summer Research Intern — UQ AI Collaboratory 12/2021 — 04/2022
Explain my AI Model: Interpretable Deep Learning for Medical Imaging — Dr Alina Bialkowski

Assessed interpretability techniques for medical-imaging models: applied transfer learning with pre-trained convolutional networks to chest X-ray classification, and used dimensionality reduction, uncertainty measures and visualisation to probe model predictions.

TEACHING

Academic Teaching Assistant 07/2022 — present
The University of Queensland, Brisbane, Australia

Head TA for Advanced Techniques for High Dimensional Data (INFS4205) and for thesis projects with a cohort of over 30 masters students (DATA7901/3, REIT7842). Tutored the practical and tutorial components of Introduction to Data Science (DATA7001).

SERVICE — REVIEWING

- 2026 KDD, AACL, ICLR, ICML GOLD REVIEWER · TOP 25%, ARR May
- 2025 ICLR, ICML, NeurIPS, ARR May, IJCAI, IEEE Transactions on Services Computing
- 2024 CIKM, IJCAI

AWARDS & HONOURS

- 2026 ICML Gold Reviewer, top 25% of reviewers
- 2025 ICLR Spotlight, top 5.1% of submissions
- 2023 Class of 2023 Valedictorian, The University of Queensland
- 2020–23 Dean's Award for Academic Excellence, four consecutive years